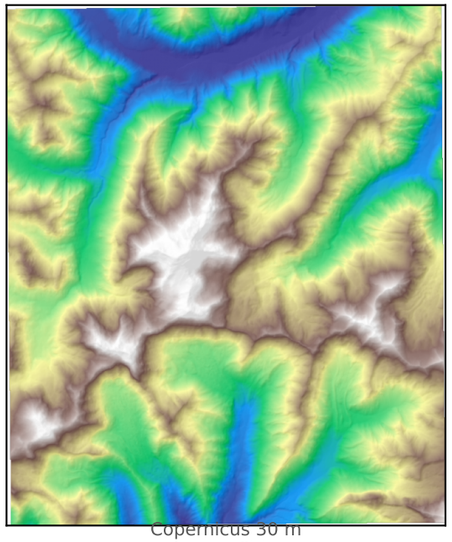
SurtGIS: High-Performance Geospatial Analysis in Rust

100+ terrain, hydrology & remote sensing algorithms · Native + WebAssembly + Python from one codebase

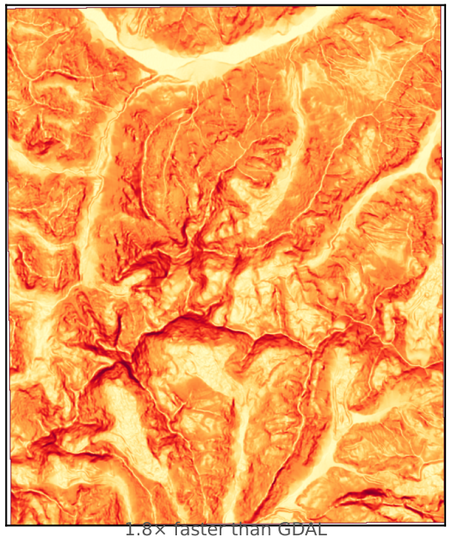
Input DEM



Copernicus 30 m
Andes, Chile

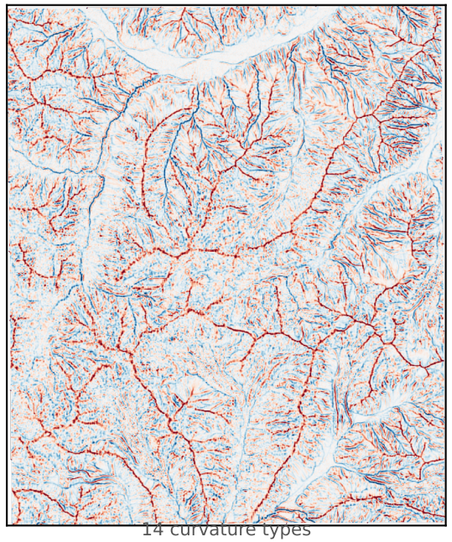
SurtGIS →

Slope



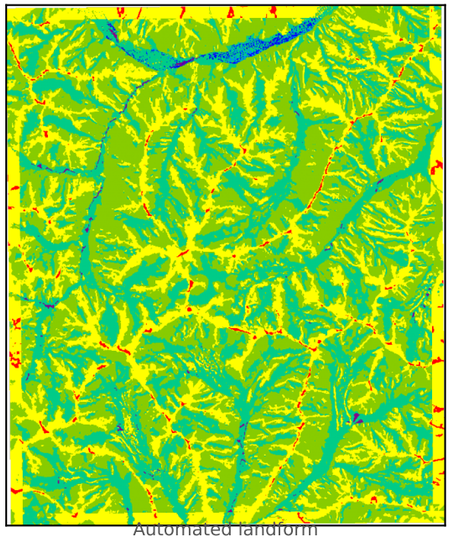
1.8× faster than GDAL
at 400M cells

Florinsky curvature



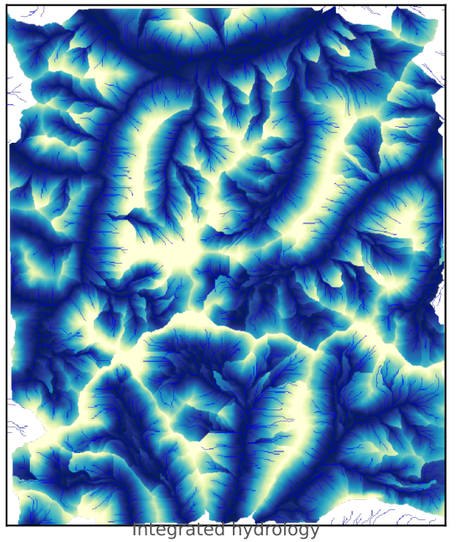
14 curvature types
open-source first

Geomorphons



Automated landform
classification

HAND + drainage



Integrated hydrology
pipeline

Desktop / Server

Native Rust + Rayon
12-core parallel

crates.io

Web Browser

WebAssembly via wasm-pack
Client-side processing

npm

Python

PyO3 bindings + NumPy
pip install surtgis

PyPI

One codebase → three deployment targets

Performance (full GeoTIFF pipeline)

